Certified Feature Selection for Wasserstein Robust Hypothesis Testing

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Abstract

We study how to select a limited number of features while designing a hypothesis test that remains reliable under class-conditional Wasserstein uncertainty. The objective is the sum of the two worst-case classwise losses. For nonnegative, nondecreasing convex losses that are finite on the real line, we obtain an exact finite optimization problem by constructing a whole-space test from its feasible solutions. The construction uses only convexity of the loss and the triangle inequality of the transport distance. We then use the risk itself to limit the coordinatewise transport prices. Eliminating the continuous variables gives a family of lower bounds: each is a residual loss divided by a transport correction. Their maximum is a convex function that agrees with the original risk at every binary feature selection. Tangents to these bounds generate a global search over feature subsets, with feasible tests providing upper bounds. Exact continuous and discrete solves terminate after finitely many subsets. The resulting certificates concern optimization within the specified uncertainty sets; population error control requires their separate calibration.

1 The selection problem

We can observe at most d of D features. The choice of features and the test using them must be optimized together: a good subset is one on which a test can keep both classwise losses small when the distributions move within their specified uncertainty sets.

Pool the observations from the two classes into $x_1, \dots, x_N \in \mathbb{R}^D$. Observation i has class $c_i \in \{0,1\}$ and weight $q_i > 0$. We encode its class by the sign $y_i = 2c_i - 1$: $y_i = -1$ for class zero and $y_i = +1$ for class one. Both c_i and y_i are fixed, observed labels, not optimization variables. Write $I_k = \{i : c_i = k\}$. The empirical distributions and uncertainty sets are

$$\widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \sum_{i \in I_k} q_i = 1, \qquad \mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \quad k = 0, 1.$$

Here δ_x denotes unit mass at x. The distance W_1 is the minimum expected L^1 distance needed to transport one distribution to the other; the distributions have finite first moments. The radii $\theta_k > 0$ and feature scales are fixed before selection. Each class has total weight one, so the pooled weights sum to two.

For an integer budget $0 \le d \le D$, a binary vector z selects $S(z) = \{j : z_j = 1\}$, with $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. A measurable score g depends only on the selected coordinates. A positive score favors class zero; the losses are therefore $\Phi(-g)$ for class zero and $\Phi(g)$ for class one. The problem is

$$\begin{aligned} V(z) &= \inf_{g = g(x_{S(z)})} \left\{ \sup_{P \in \mathcal{U}_0} \mathbb{E}_P \Phi(-g) + \sup_{P \in \mathcal{U}_1} \mathbb{E}_P \Phi(g) \right\}, \\ \mathrm{OPT} &= \min_{z \in \mathcal{Z}_d} V(z). \end{aligned} \tag{1}$$

Thus $V(z)$ already includes the best possible test for the chosen features; selection does not hold a previously fitted test fixed.

The function $\Phi : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing. Nondecreasing means that a larger score against the true class cannot reduce its loss. We consider nonconstant losses with $\Phi(0) > 0$. If $\Phi(0) = 0$ or the loss is constant, the constant test already solves the problem. Specific formulas for Φ are not needed for the construction below; examples are given in Section 6. The constant score $g = 0$ provides the common benchmark

$$0 \le V(z) \le B := 2\Phi(0).$$

2 A finite problem that still designs the whole test

The same pooled observations serve as candidate target locations. To compare source i with target ℓ, use one vector of coordinatewise distances:

$$\mathbf{d}_{i\ell} = |x_i - x_\ell| \in \mathbb{R}^D_+, \qquad d_z(x_i, x_\ell) = \mathbf{d}_{i\ell}^\top z.$$

Absolute value is taken coordinatewise. The vector stores the geometry once; its inner product with z is the distance using the selected features.

Let u_ℓ be the score assigned to target ℓ. Let a_i be a loss allowance at source i, and let $\lambda_k \geq 0$ be the price per unit of transport for class k. Write $q = (q_i)$, $\theta = (\theta_0, \theta_1)^\top$, $\lambda = (\lambda_0, \lambda_1)^\top$, and $J(a, \lambda) = q^\top a + \theta^\top \lambda$.

Theorem 1 (Exact finite formulation)**.** *For every binary z,*

$$\begin{aligned} V(z) = \min_{a,u,\lambda} \quad & J(a,\lambda) \\ \textit{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z, \qquad i, \ell = 1, \dots, N, \\ & \lambda \geq 0, \qquad a, u \in \mathbb{R}^N. \end{aligned} \tag{2}$$

The minimum is attained. Every feasible solution yields a measurable test on the entire observation space, using only the selected coordinates, with robust risk at most $J(a, \lambda)$.

The inequality in (2) has a direct meaning: after moving from source i to target ℓ, the loss is at most its source allowance plus the transport price. The source retains its class c_i and sign y_i, while u_ℓ is the score at the target. Explicitly,

$$\Phi(y_i u_\ell) = \begin{cases} \Phi(-u_\ell), & c_i = 0, \\ \Phi(u_\ell), & c_i = 1. \end{cases}$$

Thus the loss uses the source class even when the target observation originally belongs to the other class; either class can send mass to any target location.

The whole-space conclusion is essential. It allows a finite feasible objective to be used as an upper bound for the original test problem. Its proof blends two source scores, one from each class. Convexity controls the blended losses, and the triangle inequality controls the required transport allowances. The construction and both directions of the equality are given in Appendix A; no particular inverse or closed form of Φ is used.

3 Use the risk to control which coordinates are available

Taking $\ell = i$ in (2) gives $a_i \geq \Phi(y_i u_i) \geq 0$. Consequently,

$$\theta^\top \lambda \leq J(a, \lambda). \tag{3}$$

The transport-price contribution is part of the total risk, so it cannot exceed that risk. This inequality supplies the bound needed to connect feature selection with continuous test design.

Introduce two vectors $w_0, w_1 \in \mathbb{R}^D_+$ of coordinate prices. On a binary selection we can take $w_{kj} = \lambda_k z_j$: an unobserved coordinate has price zero. To obtain a useful continuous extension, use the risk bound η to constrain these prices:

$$\begin{aligned} v(z) = \min_{\eta,a,u,\lambda,w_0,w_1} \quad & \eta \\ \text{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i}, \qquad \forall i, \ell, \\ & J(a,\lambda) \leq \eta, \qquad \lambda \geq 0, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0, 1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq \eta z. \end{aligned} \tag{4}$$

Vector inequalities hold coordinatewise; a, u, η are real variables. The first line designs the test using the coordinate prices. The next lines bound its risk and keep each coordinate price below the class price. The last line makes an unselected coordinate unavailable and ties the allowed prices to the risk being minimized.

For binary z, an optimizer of (2) enters (4) with $\eta = J$ and $w_k = \lambda_k z$, by (3). Conversely, a feasible point of (4) has zero prices on unselected coordinates and $w_k \le \lambda_k z$. The same a, u, λ therefore satisfy (2), with objective at most η. Hence $v(z) = V(z)$ at every binary selection. For fractional $z \in [0,1]^D$, (4) defines the extension used to construct lower bounds.

Fixing z makes (4) a convex problem. This alone does not prove that its value $v(z)$ is convex in z, since the last line contains the product ηz. The next step establishes that property and supplies information that remains useful when the feature subset changes.

4 One solved subset gives lower bounds for all subsets

Consider a nonnegative matrix Π with $\Pi_{i\ell}$ units of mass sent from source i to target ℓ. Its rows retain the source weights, $\Pi\mathbf{1} = q$. The masses received from the two classes and their coordinatewise movements are

$$p_{k\ell} = \sum_{i \in I_k} \Pi_{i\ell}, \qquad A_k(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} \mathbf{d}_{i\ell} \in \mathbb{R}^D_+.$$

At a target receiving both classes, one score must serve both. The least weighted loss that this common score can achieve is

$$\Psi(p_0, p_1) = \inf_s \{p_0 \Phi(-s) + p_1 \Phi(s)\}, \qquad H(\Pi) = \sum_\ell \Psi(p_{0\ell}, p_{1\ell}). \tag{5}$$

The infimum permits limiting scores when one class has zero mass. H measures the loss left after the best score is chosen at every target. It is nonnegative. Scaling all masses by $t \ge 0$ scales H by t; this property concerns the masses and does not require Φ itself to be homogeneous.

To use a matrix that may move more mass than the original budget allows, introduce a nonnegative correction vector $\mu \in \mathbb{R}^D_+$. The pair (Π, μ) must satisfy

$$\Pi \ge 0, \qquad \Pi\mathbf{1} = q, \qquad \mu \ge 0, \qquad \mathbf{1}^\top \big(A_k(\Pi) - \theta_k \mu\big)_+ \le \theta_k, \quad k = 0, 1. \tag{6}$$

Here $t_+ = \max(t, 0)$, applied coordinatewise to a vector. After $\theta_k \mu_j$ is deducted from movement in coordinate j, the remaining positive movements must fit inside the class budget θ_k. These conditions contain no z: the same pair can be used at any selection. The matrix need not be a transport plan inside the original uncertainty balls. The correction is what makes it useful as a risk lower bound.

The reason for the correction in the denominator can be seen directly. For $0 \le z \le 1$, (6) implies $z^\top A_k(\Pi) \le \theta_k(1 + \mu^\top z)$. For any feasible point of (2) on a binary selection,

$$\begin{aligned} H(\Pi) &\le \sum_{i,\ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ &\le q^\top a + \sum_k \lambda_k z^\top A_k(\Pi) \\ &\le J(a, \lambda) + (\mu^\top z) \theta^\top \lambda \\ &\le (1 + \mu^\top z) J(a, \lambda). \end{aligned}$$

The four steps use, respectively, best-score loss, finite loss constraints, the corrected transport budgets, and (3). Thus $H(\Pi)/(1 + \mu^\top z)$ is a lower bound on $V(z)$.

Theorem 2 (Exact representation and convexity)**.** *For every* $z \in [0,1]^D$,

$$v(z) = \max_{\Pi,\mu \text{ satisfying (6)}} \frac{H(\Pi)}{1+\mu^\top z}. \tag{7}$$

The maximum is attained, v *is convex, and* $v(z) = V(z)$ *on binary selections.*

The preceding inequalities explain why each fraction is a valid lower bound. Equality with v requires more: minimizing the Lagrangian of (4) gives the row masses, transport inequalities and one normalization equation. Rescaling those multipliers gives exactly (7). Appendix B derives the objective and every constraint by eliminating one group of variables at a time. For fixed (Π, μ), the fraction is convex because its numerator is nonnegative and its denominator is a positive affine function. The maximum of these convex lower functions is therefore convex.

5 From lower bounds to a global algorithm

At a selection z^t, solving (4) and its dual gives a pair (Π^t, μ^t). Write $h_t = H(\Pi^t)$ for its remaining target loss and $r_t = 1 + (\mu^t)^\top z^t$ for the correction at the current selection. The function $f_t(z) = h_t/[1 + (\mu^t)^\top z]$ is a lower bound everywhere. Its gradient at z^t is $-h_t\mu^t/r_t^2$, so its tangent is

$$L_t(z) = \frac{h_t}{r_t} - \frac{h_t}{r_t^2}(\mu^t)^\top(z - z^t) \le v(z). \tag{8}$$

This affine inequality is the global cut. Convexity puts the tangent below f_t, and (7) puts f_t below v. An optimal pair makes $L_t(z^t) = v(z^t)$. A merely feasible pair also gives a valid cut, using its own h_t. This argument does not assume that v is differentiable.

Given the accumulated cuts, solve the discrete master problem

$$\min_{z \in \mathcal{Z}_d, \eta} \eta \qquad \text{subject to} \qquad \eta \ge 0, \quad \eta \ge L_t(z) \text{ for every stored cut.} \tag{9}$$

It searches the entire set of allowed feature subsets. Each new cut changes the lower model for all subsets, while a continuous solve refits the entire test at the next candidate. The search is not confined to a neighborhood of the current subset or to a fixed combination of earlier tests.

Algorithm 1 Global feature selection

Require: Initial $z^0 \in \mathcal{Z}_d$; tolerance $\varepsilon \ge 0$.

1: Set $t = 0$, LB $= 0$, UB $= B$; retain z^0 and $g = 0$ as the best feasible test.
2: **while** UB $-$ LB $> \varepsilon$ **do**
3: At the current binary selection z^t, solve the full problem (4).
4: Construct an original feasible test with bound $b_t = J(a^t, \lambda^t)$. If $b_t <$ UB, update UB and retain this selection and test.
5: Obtain a feasible dual pair and store the cut (8) using its own numerator h_t; retain every previous cut.
6: Solve (9); obtain a valid global lower bound ℓ_t and binary candidate z^{t+1}; set LB $\leftarrow \max\{\text{LB}, \ell_t\}$ and $t \leftarrow t + 1$.
7: If the time or accuracy budget is exhausted, return the best test and the achieved bounds.
8: **end while**
9: **return** Best feature subset and test, with interval [LB, UB].

Theorem 3 (Global validity and finite termination)**.** *For a pair satisfying* (6)*, its cut* (8) *is a lower bound on v throughout $[0,1]^D$ and on V throughout $\mathcal{Z}_d$. The master lower bounds and the feasible-test upper bounds in Algorithm 1 satisfy*

$$\mathrm{LB} \leq \mathrm{OPT} \leq \mathrm{UB}. \tag{10}$$

If no budget stop occurs, exact continuous solutions, optimal dual pairs and globally solved masters require at most $|\mathcal{Z}_d|$ continuous evaluations. For $\varepsilon = 0$, the final bounds equal OPT*. Any stop with* $\mathrm{UB} - \mathrm{LB} \leq \varepsilon$ *guarantees that the returned test has risk at most* $\mathrm{OPT} + \varepsilon$*.*

The mechanism is that a solved subset has both a touching cut and an equally good feasible test. If an exact master selects it again, the bounds coincide and the algorithm stops before reevaluating it. Every continuing iteration therefore evaluates a new subset. Appendix C proves Theorem 3, from global cut validity to the bound invariant and finite termination. It also gives sufficient accuracy conditions for finite termination with inexact solves. Feasibility alone preserves valid bounds. An interrupted master's global solver bound is usable; its current feasible objective is not a lower bound.

Adding available coordinates cannot worsen V, because a test can ignore them. The search may therefore use $\mathbf{1}^\top z = d$ when features have no additional charge. Cuts from fractional selections may strengthen the master, but only binary feasible tests update the original upper bound.

6 Loss choices and the scope of the result

The loss enters through Φ in the continuous problem and through Ψ in the dual objective. Examples covered by the same derivation are shown below; boundary values use continuous limits.

Loss	$\Phi(s)$	$\Psi(p_0, p_1)$
Half hinge	$\frac{1}{2}(1+s)_+$	$\min(p_0, p_1)$
Logistic	$\log(1+e^s)$	$(p_0+p_1)\log(p_0+p_1) - p_0 \log p_0 - p_1 \log p_1$
Exponential	e^s	$2\sqrt{p_0 p_1}$
Squared hinge	$(1+s)_+^2$	$4p_0p_1/(p_0+p_1)$

For example, half-hinge masses $p_0 = 0.2$ and $p_1 = 0.5$ leave loss 0.2: the best common score favors the larger mass but cannot distinguish the smaller mass at the same location. Appendix D derives all four formulas. For half hinge, a score clipped to $[-1, 1]$ gives randomized class-zero probability $(1 + g)/2$, so $V/2$ is the equal-prior worst-case error. Other losses define surrogate objectives. The current implementation uses half hinge and logistic; the other rows illustrate the theoretical scope.

The loss formulas need not be fixed before deriving the algorithm. The transport geometry still matters: the whole-space construction uses the triangle inequality, and the coordinate-price formulation uses the L^1 decomposition. The certified interval concerns the stated robust optimization problem; whether its uncertainty sets cover an unknown population is a separate statistical question.

A Proof of Theorem 1: the finite formulation

A.1 From distributions to finite transports

Fix a binary selection z and let $S = \{j : z_j = 1\}$. For any two points, the distance seen by the test is

$$d_z(x, x') := \sum_{j=1}^{D} z_j |x_j - x'_j| = \|x_S - x'_S\|_1 \le \|x - x'\|_1. \tag{A.1}$$

Write P_S for the distribution of x_S under P. Projection cannot increase transport cost. Conversely, mass from source x_i can be sent to the selected coordinates of target x_ℓ while retaining its unselected coordinates:

$$\begin{aligned} W_1(P_S, \widehat{P}_{k,S}) &\le W_1(P, \widehat{P}_k), \\ \|x_i - (x_{\ell,S}, x_{i,S^c})\|_1 &= d_z(x_i, x_\ell), \\ g(x_{\ell,S}, x_{i,S^c}) &= g(x_\ell). \end{aligned} \tag{A.2}$$

The last identity holds because g uses only S. Retain each source identity when lifting: coincident projected sources split their transported mass according to their weights. Replacing $x_{\ell,S}$ by an arbitrary projected target gives the same lifting for general projected distributions, with the same cost and finite first moments.

For a score vector $u = (u_\ell)_{\ell=1}^N$, let $Q_k(u)$ be the largest class-k loss obtained by sending mass to the projected pool. Here $\Pi_{i\ell}$ is the mass from source $i \in I_k$ to target ℓ; this matrix has only class-k source rows:

$$\begin{aligned} Q_k(u) = \max_{\Pi \ge 0} \quad & \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ \text{subject to} \quad & \sum_{\ell} \Pi_{i\ell} = q_i \quad (i \in I_k), \\ & \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \le \theta_k. \end{aligned} \tag{A.3}$$

Staying at the source is feasible, and $0 \le \Pi_{i\ell} \le q_i$ makes the feasible set compact.

To derive its dual, use unrestricted multipliers a_i for the row equalities and $\lambda_k \ge 0$ for the cost budget. For this maximization problem the Lagrangian is

$$\begin{aligned} \mathcal{L}_k &= \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) + \sum_{i \in I_k} a_i \left(q_i - \sum_{\ell} \Pi_{i\ell} \right) \\ &\quad + \lambda_k \left(\theta_k - \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \right) \\ &= \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k + \sum_{i \in I_k, \ell} \Pi_{i\ell} [\Phi(y_i u_\ell) - a_i - \lambda_k d_z(x_i, x_\ell)]. \end{aligned} \tag{A.4}$$

The row and budget constraints have been moved into the Lagrangian; only $\Pi \ge 0$ remains in its supremum. For a primal feasible transport, the row terms vanish and the budget term is nonnegative, so $\mathcal{L}_k$ is an upper bound on its loss. A positive coefficient of any $\Pi_{i\ell}$ makes the supremum infinite. If all coefficients are nonpositive, setting $\Pi = 0$ attains that supremum. Thus

$$\sup_{\Pi \ge 0} \mathcal{L}_k = \begin{cases} \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k, & \Phi(y_i u_\ell) \le a_i + \lambda_k d_z(x_i, x_\ell) \quad (i \in I_k,\ \forall \ell), \\ +\infty, & \text{otherwise.} \end{cases} \tag{A.5}$$

Minimizing this upper bound gives the dual; finite linear programming strong duality gives equality:

$$Q_k(u) = \min_{\substack{a_i \in \mathbb{R},\ \lambda_k \geq 0 \\ \Phi(y_i u_\ell) \leq a_i + \lambda_k d_z(x_i, x_\ell)\ (i \in I_k,\ \forall \ell)}} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right]. \tag{A.6}$$

Let $T(z)$ be the infimum of (2). Its variables u are shared by the two classes, so (A.6) gives $T(z) = \inf_u [Q_0(u) + Q_1(u)]$. For any original test, define its induced scores $u(g) = (g(x_1), \ldots, g(x_N))$. Lift each feasible transport in (A.3) using (A.2). The resulting full-space distribution lies in $\mathcal{U}_k$, and its class loss is precisely $\sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i g(x_\ell))$. Consequently, for every such g,

$$Q_k(u(g)) \leq \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g). \tag{A.7}$$

Restricting the infimum to induced score vectors and then using this pointwise inequality gives

$$\begin{aligned} T(z) &= \inf_u [Q_0(u) + Q_1(u)] \\ &\leq \inf_g [Q_0(u(g)) + Q_1(u(g))] \leq V(z). \end{aligned} \tag{A.8}$$

Separate scores at coincident projected targets can only relax the first infimum. No interchange of a function-space infimum and a distributional supremum is used.

A.2 Constructing a test on the whole space

Take any feasible (a, u, λ) in (2). We seek a score whose class-k loss at every x is below every source allowance $a_i + \lambda_k d_z(x_i, x)$, $i \in I_k$. The smallest such allowance is

$$\begin{aligned} E_k(x) &= \min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}, \\ i_k(x) &= \min \arg\min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}. \end{aligned} \tag{A.9}$$

The second line chooses the smallest source index in a tie. Fix x and abbreviate $i_k = i_k(x)$. Let c be the distance between the chosen sources, and r_k their distances to x:

$$c = d_z(x_{i_0}, x_{i_1}), \qquad r_k = d_z(x_{i_k}, x), \qquad c \leq r_0 + r_1. \tag{A.10}$$

For $c > 0$, blend their scores using

$$\alpha = \frac{\min(c, r_0)}{c}, \qquad g(x) = (1-\alpha) u_{i_0} + \alpha u_{i_1}. \tag{A.11}$$

This choice keeps the two transport allowances within the available distances:

$$\begin{aligned} 0 \leq \alpha \leq 1, \qquad & \alpha c = \min(c, r_0) \leq r_0, \\ & (1-\alpha)c = c - \min(c, r_0) = \max(c - r_0, 0) \leq r_1. \end{aligned} \tag{A.12}$$

The final inequality follows from $c \leq r_0 + r_1$ and $r_1 \geq 0$.

In (2), substitute source–target pairs (i_0, i_0), (i_0, i_1), (i_1, i_0), and (i_1, i_1), respectively. The first source has sign -1, the second has sign $+1$, and their own-source distances are zero. These four substitutions give

$$\begin{aligned} \Phi(-u_{i_0}) &\leq a_{i_0}, & \Phi(-u_{i_1}) &\leq a_{i_0} + \lambda_0 c, \\ \Phi(u_{i_0}) &\leq a_{i_1} + \lambda_1 c, & \Phi(u_{i_1}) &\leq a_{i_1}. \end{aligned} \tag{A.13}$$

Apply convexity separately to the two class losses, then insert these endpoint bounds and (A.12):

$$\begin{aligned}\Phi(-g(x)) &\le (1-\alpha)\Phi(-u_{i_0}) + \alpha\Phi(-u_{i_1}) \\ &\le (1-\alpha)a_{i_0} + \alpha(a_{i_0} + \lambda_0 c) = a_{i_0} + \lambda_0 \alpha c \\ &\le a_{i_0} + \lambda_0 r_0 = E_0(x), \\ \Phi(g(x)) &\le (1-\alpha)\Phi(u_{i_0}) + \alpha\Phi(u_{i_1}) \\ &\le (1-\alpha)(a_{i_1} + \lambda_1 c) + \alpha a_{i_1} = a_{i_1} + \lambda_1(1-\alpha)c \\ &\le a_{i_1} + \lambda_1 r_1 = E_1(x).\end{aligned} \tag{A.14}$$

If $c = 0$, define $g(x) = u_{i_0}$ instead. The first and third inequalities in (A.13) directly give

$$\Phi(-g(x)) \le a_{i_0} \le E_0(x), \qquad \Phi(g(x)) \le a_{i_1} + \lambda_1 c = a_{i_1} \le E_1(x). \tag{A.15}$$

Finite comparisons of continuous functions make $i_k(x)$ Borel measurable; the piecewise construction then makes g Borel measurable. It uses only x_S and lies between $\min_\ell u_\ell$ and $\max_\ell u_\ell$. It extends the loss bounds, without needing to interpolate every prescribed u_ℓ.

A.3 Integrating the source allowances

Fix $P \in \mathcal{U}_k$ and choose a coupling of cost at most θ_k. Retain the source indices, splitting coincident atoms according to their weights. Write γ_k for the resulting measure on $I_k \times \mathbb{R}^D$, with target marginal P. Its rows and projected cost satisfy

$$\gamma_k(\{i\} \times \mathbb{R}^D) = q_i, \qquad \int d_z(x_i, x)\, d\gamma_k(i,x) \le \int \|x_i - x\|_1\, d\gamma_k(i,x) \le \theta_k. \tag{A.16}$$

By construction, $\Phi((2k-1)g(x)) \le E_k(x) \le a_i + \lambda_k d_z(x_i, x)$ for every source $i \in I_k$. The two marginals of γ_k therefore give

$$\begin{aligned}\mathbb{E}_P \Phi((2k-1)g) &= \int \Phi((2k-1)g(x))\, d\gamma_k(i,x) \\ &\le \int a_i\, d\gamma_k(i,x) + \lambda_k \int d_z(x_i,x)\, d\gamma_k(i,x) \\ &= \sum_{i \in I_k} a_i \gamma_k(\{i\} \times \mathbb{R}^D) + \lambda_k \int d_z(x_i,x)\, d\gamma_k(i,x) \\ &= \sum_{i \in I_k} q_i a_i + \lambda_k \int d_z(x_i,x)\, d\gamma_k(i,x) \\ &\le \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k.\end{aligned} \tag{A.17}$$

The right side does not depend on P. Taking the supremum in each class and adding gives

$$\sum_{k=0}^{1} \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g) \le \sum_{k=0}^{1} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right] = J(a, \lambda).$$

Thus every finite feasible point yields an original feasible test with risk at most J. Taking the infimum over those points gives $V(z) \le T(z)$; together with (A.8), this proves Theorem 1's value equality and test construction.

A.4 Attainment of the primal minima

We now prove attainment in Theorem 1, and also the primal attainment needed for Theorem 2. Because Φ is nonconstant and nondecreasing, some $s_1 < s_2$ satisfy $\Phi(s_2) > \Phi(s_1)$. Define $b = [\Phi(s_2) - \Phi(s_1)]/(s_2 - s_1) > 0$. Convexity gives, for $t > s_2$,

$$\frac{\Phi(t) - \Phi(s_2)}{t - s_2} \geq b \quad \Longrightarrow \quad \Phi(t) \geq \Phi(s_2) + b(t - s_2) \longrightarrow \infty. \tag{A.18}$$

The point $u = 0$, $a_i = \Phi(0)$ and $\lambda = 0$ is feasible in (2), with $J = B$. Adding $w_0 = w_1 = 0$ and $\eta = B$ makes it feasible in (4) for every $z \in [0,1]^D$. Thus both feasible sets with objective at most B are nonempty. On this objective range, the own-source constraints and nonnegativity give

$$\begin{aligned} 0 \leq q_i a_i \leq q^\top a \leq J \leq B \quad &\Longrightarrow\ 0 \leq a_i \leq B/q_i, \\ 0 \leq \theta_k \lambda_k \leq \theta^\top \lambda \leq J \leq B \quad &\Longrightarrow\ 0 \leq \lambda_k \leq B/\theta_k, \\ 0 \leq w_{kj} \leq \lambda_k \leq B/\theta_k, \quad &0 \leq J \leq \eta \leq B \quad \text{in (4).} \end{aligned} \tag{A.19}$$

Fix one source $i_k \in I_k$ for each class, and define the finite constant

$$C = \max_{k,\ell} \left\{ B/q_{i_k} + (B/\theta_k) \|\mathbf{d}_{i_k \ell}\|_1 \right\}.$$

In either primal model, nonnegative coordinate distances imply

$$\Phi((2k-1)u_\ell) \leq B/q_{i_k} + (B/\theta_k) \|\mathbf{d}_{i_k \ell}\|_1 \leq C.$$

Choose $R > 0$ with $\Phi(R) > C$, possible by (A.18). Then

$$u_\ell \geq R \implies \Phi(u_\ell) > C, \qquad u_\ell \leq -R \implies \Phi(-u_\ell) > C; \qquad |u_\ell| < R.$$

All variables are bounded; continuity of the finite convex loss makes these feasible sets closed. Their nonempty compactness gives both primal minima.

B Proof of Theorem 2: the dual representation

B.1 Multipliers and the Lagrangian

Fix $z \in [0,1]^D$ in (4). The variables η, a, u are free, whereas λ, w_0, w_1 remain nonnegative. Write each remaining constraint with left side at most zero and attach the multiplier shown below:

Constraint	Nonnegative multiplier
$\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} \leq 0$	$\bar{\Pi}_{i\ell}$
$J - \eta \leq 0$	τ
$w_{kj} - \lambda_k \leq 0$	$\bar{\beta}_{kj}$
$\theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j \leq 0$	$\bar{\mu}_j$

Here $\bar{\Pi} \in \mathbb{R}_+^{N \times N}$, $\tau \in \mathbb{R}_+$, and $\bar{\beta}_k, \bar{\mu} \in \mathbb{R}_+^D$. The bars distinguish these multipliers from the row-normalized matrix and correction vector in the main text.

Adding multiplier times constraint to the objective gives

$$\begin{aligned} \mathcal{L} = \eta + \tau(J - \eta) + \sum_{i,\ell} \bar{\Pi}_{i\ell} [\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i}] \\ + \sum_k \bar{\beta}_k^\top (w_k - \lambda_k \mathbf{1}) + \bar{\mu}^\top (\theta_0 w_0 + \theta_1 w_1 - \eta z). \end{aligned} \tag{B.1}$$

At a feasible point of (4), every added constraint term is nonpositive. Therefore, with the multipliers fixed,

$$\inf_{\substack{\eta\in\mathbb{R},\ a,u\in\mathbb{R}^N\\ \lambda,w_0,w_1\geq 0}} \mathcal{L} \leq \mathcal{L}(\eta, a, u, \lambda, w_0, w_1) \leq \eta.$$

The infimum is a lower bound on $v(z)$. The dual problem maximizes this lower bound over the multipliers; equality will be verified below. All four constraint groups are now in $\mathcal{L}$; they are not imposed again when taking its infimum. In particular, the nonnegativity of a and η implied by the original constraints does not restrict their domains in this infimum.

To collect the score and price terms, define

$$\bar{p}_{k\ell} = \sum_{i\in I_k} \bar{\Pi}_{i\ell}, \qquad A_k(\bar{\Pi}) = \sum_{i\in I_k,\ell} \bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}. \tag{B.2}$$

The first quantity is the mass of class k at target ℓ; the second is its vector of coordinate movements. The regrouping uses

$$\begin{aligned}
\sum_{i,\ell} \bar{\Pi}_{i\ell} a_i &= \sum_i a_i \sum_\ell \bar{\Pi}_{i\ell},\\
\sum_{i,\ell} \bar{\Pi}_{i\ell} \Phi(y_i u_\ell) &= \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)],\\
\sum_{i,\ell} \bar{\Pi}_{i\ell} \mathbf{d}_{i\ell}^\top w_{c_i} &= \sum_k w_k^\top A_k(\bar{\Pi}),\\
\tau J &= \sum_i \tau q_i a_i + \sum_k \tau\theta_k\lambda_k.
\end{aligned} \tag{B.3}$$

Thus each primal variable group has its own term:

$$\begin{aligned}
\mathcal{L} = \eta(1-\tau-\bar{\mu}^\top z) &+ \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell})\\
&+ \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)]\\
&+ \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top\bar{\beta}_k) + \sum_k w_k^\top[\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})].
\end{aligned} \tag{B.4}$$

B.2 Minimizing over the primal variables

Keep all multipliers fixed. The dual value is the infimum of (B.4) over $\eta \in \mathbb{R}$, $a, u \in \mathbb{R}^N$ and $\lambda, w_0, w_1 \geq 0$. Since these domains are independent, minimize each term separately.

For a linear coefficient b, the two rules are

$$\inf_{t\in\mathbb{R}} bt = \begin{cases} 0, & b = 0,\\ -\infty, & b \neq 0, \end{cases} \qquad \inf_{t\geq 0} bt = \begin{cases} 0, & b \geq 0,\\ -\infty, & b < 0. \end{cases}$$

For the first rule, $t = -nb$ gives $bt = -nb^2 \to -\infty$ when $b \neq 0$. For the second, $t = n$ gives $bt \to -\infty$ when $b < 0$; otherwise the minimum is zero at $t = 0$.

The risk variable and source allowances. Applying the free-variable rule gives

$$\begin{aligned}
\inf_{\eta\in\mathbb{R}} \eta(1-\tau-\bar{\mu}^\top z) > -\infty &\iff 1-\tau-\bar{\mu}^\top z = 0,\\
\inf_{a\in\mathbb{R}^N} \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell}) > -\infty &\iff \tau q_i - \sum_\ell \bar{\Pi}_{i\ell} = 0 \quad (\forall i).
\end{aligned} \tag{B.5}$$

Both terms then contribute zero. The second set of equalities is $\bar{\Pi}\mathbf{1} = \tau q$.

The scores. One score u_ℓ serves the two masses at target ℓ. By the definition of Ψ in (5),

$$\begin{aligned}\inf_{u\in\mathbb{R}^N} \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)] &= \sum_\ell \inf_{s\in\mathbb{R}} [\bar{p}_{0\ell}\Phi(-s) + \bar{p}_{1\ell}\Phi(s)] \\ &= \sum_\ell \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) = H(\bar{\Pi}).\end{aligned} \tag{B.6}$$

To justify the first equality even at boundary masses, fix $\varepsilon > 0$ and choose each u_ℓ so that

$$\Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) \le \bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell) \le \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) + \varepsilon/N.$$

Summing puts the loss between $H(\bar{\Pi})$ and $H(\bar{\Pi}) + \varepsilon$; let $\varepsilon \downarrow 0$. The infima are finite because

$$0 \le \Psi(p_0, p_1) \le \Phi(0)(p_0 + p_1), \qquad 0 \le H(\bar{\Pi}) \le \Phi(0) \sum_{i,\ell} \bar{\Pi}_{i\ell} = \tau B.$$

The last equality uses the row condition and $\sum_i q_i = 2$.

The class and coordinate prices. Applying the nonnegative-variable rule gives

$$\begin{aligned}\inf_{\lambda\ge 0} \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top \bar{\beta}_k) > -\infty \iff \mathbf{1}^\top\bar{\beta}_k \le \tau\theta_k \quad (\forall k), \\ \inf_{w_0, w_1 \ge 0} \sum_k w_k^\top [\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})] > -\infty \iff A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k\bar{\mu} \quad (\forall k).\end{aligned} \tag{B.7}$$

Each term contributes zero when its condition holds. Therefore the minimized Lagrangian is $H(\bar{\Pi})$ exactly under (B.5) and (B.7), and is $-\infty$ otherwise. Maximizing this finite value over nonnegative multipliers is the dual problem.

It remains to remove $\bar{\beta}_k$. Coordinatewise, its conditions imply

$$\begin{aligned}\bar{\beta}_{kj} \ge 0, \quad A_{kj}(\bar{\Pi}) \le \bar{\beta}_{kj} + \theta_k\bar{\mu}_j \iff \bar{\beta}_{kj} \ge \max(0, A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j), \\ \mathbf{1}^\top\bar{\beta}_k \le \tau\theta_k \implies \sum_j (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+ \le \tau\theta_k.\end{aligned} \tag{B.8}$$

Conversely, if the last inequality holds, choose

$$\bar{\beta}_k = (A_k(\bar{\Pi}) - \theta_k\bar{\mu})_+.$$

This vector is nonnegative, has sum at most $\tau\theta_k$, and satisfies $A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k\bar{\mu}$. Thus both directions are exact, and the dual is

$$\begin{aligned}\max_{\tau,\bar{\Pi},\bar{\mu}\ge 0} \quad & H(\bar{\Pi}) \\ \text{subject to} \quad & \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = 1, \\ & \sum_j (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+ \le \tau\theta_k, \quad k = 0, 1.\end{aligned} \tag{B.9}$$

We next verify equality with the primal optimum and attainment, before dividing by τ.

B.3 Equality of primal and dual values

Primal attainment was established in Appendix A.4.

The restriction to objective values at most B was used only to prove existence; it is not added to (4). For strong duality in that program, first identify the prices already fixed by a zero coordinate:

$$z_j = 0 \implies 0 \le \theta_0 w_{0j} + \theta_1 w_{1j} \le \eta z_j = 0 \implies w_{0j} = w_{1j} = 0.$$

This uses $\theta_k > 0$ and $w_{kj} \ge 0$. Delete these fixed variables and the resulting tautological coordinate inequalities. On the remaining coordinates choose

$$u = 0, \quad a_i = \Phi(0) + 1, \quad \lambda_k = 1, \quad w_k = \delta z, \quad 0 < \delta < 1, \quad \eta = J + 1 + \delta(\theta_0 + \theta_1). \tag{B.10}$$

Here $J = 2[\Phi(0) + 1] + \theta_0 + \theta_1$. Every relevant residual is strictly negative:

$$\begin{aligned} \Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} &= -1 - \delta \mathbf{d}_{i\ell}^\top z < 0, \\ w_{kj} - \lambda_k &= \delta z_j - 1 < 0, \\ J - \eta &= -1 - \delta(\theta_0 + \theta_1) < 0, \\ \theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j &= -(J+1) z_j < 0 \quad (z_j > 0). \end{aligned} \tag{B.11}$$

Also $w_{kj} > 0$ on those coordinates and $\lambda_k > 0$. Finite-dimensional convex strong duality applies to this reduced problem: the dual maximum is attained and equals $v(z)$.

To restore an omitted coordinate, use

$$z_j = 0 : \quad \bar{\beta}_{kj} = 0, \qquad \bar{\mu}_j = \max_k A_{kj}(\bar{\Pi})/\theta_k.$$

Then $\bar{\beta}_{kj} + \theta_k \bar{\mu}_j \ge A_{kj}(\bar{\Pi})$ for both classes, no allocation budget increases, and $\bar{\mu}_j z_j = 0$ leaves normalization unchanged. Thus the full dual (B.9) also has an attained maximum equal to $v(z)$.

B.4 Removing the normalization

We first justify division by τ. If $\tau = 0$, nonnegativity and the row condition give

$$\bar{\Pi}\mathbf{1} = 0 \implies \bar{\Pi} = 0 \implies H(\bar{\Pi}) = \sum_\ell \inf_s 0 = 0.$$

For a positive scale, the mass-scaling identity follows directly from the infimum:

$$\begin{aligned} \Psi(\tau p_0, \tau p_1) &= \inf_s \tau[p_0 \Phi(-s) + p_1 \Phi(s)] = \tau \Psi(p_0, p_1), \\ H(\tau \Pi) &= \sum_\ell \Psi(\tau p_{0\ell}, \tau p_{1\ell}) = \tau H(\Pi), \qquad \tau > 0. \end{aligned} \tag{B.12}$$

There is, however, a feasible point with strictly positive objective. Choose a target ℓ_*, and let e_{ℓ_*} be the vector with entry one at that target and zero elsewhere. Sending every source to this target means $\Pi = q e_{\ell_*}^\top$. Its received masses are one from each class, so

$$H(\Pi) = \Psi(1,1), \qquad \Phi(-s) + \Phi(s) \ge 2\Phi(0), \qquad \Psi(1,1) = 2\Phi(0) = B > 0.$$

The last equality follows by taking $s = 0$. Set

$$\mu_j = \max_k \frac{A_{kj}(\Pi)}{\theta_k}, \qquad \tau = \frac{1}{1 + \mu^\top z}, \qquad \bar{\Pi} = \tau \Pi, \qquad \bar{\mu} = \tau \mu. \tag{B.13}$$

These choices satisfy every normalized constraint:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = \tau(1 + \mu^\top z) = 1, \\ A_k(\Pi) - \theta_k \mu \le 0 \implies (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ = 0, \\ H(\bar{\Pi}) = \tau H(\Pi) = \tau B > 0. \end{aligned}$$

Thus an attained optimal normalized solution cannot have $\tau = 0$.

Now take any normalized feasible point with $\tau > 0$ and define $\Pi = \bar{\Pi}/\tau$, $\mu = \bar{\mu}/\tau$. The row and movement constraints transform as follows:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q \iff \Pi\mathbf{1} = q, \\ A_k(\bar{\Pi}) = \tau A_k(\Pi), \\ (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ = \tau (A_k(\Pi) - \theta_k \mu)_+, \\ \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \le \tau \theta_k \iff \sum_j (A_{kj}(\Pi) - \theta_k \mu_j)_+ \le \theta_k. \end{aligned} \tag{B.14}$$

The third line uses $(\tau t)_+ = \tau t_+$ for $\tau > 0$. Normalization then gives the denominator and the objective:

$$\begin{aligned} \tau + \bar{\mu}^\top z = 1 \iff \tau(1 + \mu^\top z) = 1 \iff \tau = (1 + \mu^\top z)^{-1}, \\ H(\bar{\Pi}) = \tau H(\Pi) = \frac{H(\Pi)}{1 + \mu^\top z}. \end{aligned} \tag{B.15}$$

The denominator is positive since $\mu, z \ge 0$. Conversely, start with any pair satisfying (6), choose τ as in (B.15), and multiply both objects by τ. Equation (B.14) verifies the row and movement conditions, and $\tau(1 + \mu^\top z) = 1$ verifies normalization. The objectives agree in both directions. This proves (7) and its attained maximum.

B.5 Equality on binary selections

For an original finite feasible point, put $\eta = J$ and $w_k = \lambda_k z$. Then

$$\begin{aligned} 0 \le w_k \le \lambda_k \mathbf{1}, \\ \theta_0 w_{0j} + \theta_1 w_{1j} = z_j \theta^\top \lambda \le z_j J = z_j \eta, \\ a_i + \mathbf{d}_{i\ell}^\top w_{c_i} = a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z. \end{aligned} \tag{B.16}$$

The middle inequality is (3). All constraints of (4) therefore hold, giving $v(z) \le V(z)$.

Conversely, for a risk-model feasible point, $\theta_k > 0$ and nonnegative prices give

$$\begin{aligned} z_j = 0 \implies \theta_0 w_{0j} + \theta_1 w_{1j} \le 0 \implies w_{0j} = w_{1j} = 0, \\ z_j = 1 \implies w_{kj} \le \lambda_k, \\ \implies \quad w_k \le \lambda_k z. \end{aligned} \tag{B.17}$$

Since $\mathbf{d}_{i\ell} \ge 0$ coordinatewise,

$$\Phi(y_i u_\ell) \le a_i + \mathbf{d}_{i\ell}^\top w_{c_i} \le a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z.$$

Thus (a, u, λ) is feasible in (2), with $V(z) \le J \le \eta$. Taking the minimum over risk-model feasible points gives $V(z) \le v(z)$. Combining both directions yields $v(z) = V(z)$.

B.6 Convexity and completion of Theorem 2

Fix one feasible pair (Π, μ) in (6). Its remaining loss $h = H(\Pi) \ge 0$ is constant when z changes. For $z \in [0,1]^D$, define the scalar correction $r(z) = 1 + \mu^\top z > 0$ and its lower function $f(z) = h/r(z)$. For two coordinates $j, m \in \{1, \dots, D\}$, the chain rule gives

$$\begin{aligned} \frac{\partial r}{\partial z_j} &= \mu_j, \qquad \frac{\partial f}{\partial z_j} = h[-r(z)^{-2}]\mu_j = -\frac{h\mu_j}{r(z)^2}, \\ \frac{\partial^2 f}{\partial z_m \partial z_j} &= -h\mu_j[-2r(z)^{-3}]\mu_m = \frac{2h\mu_j\mu_m}{r(z)^3}, \\ \nabla^2 f(z) &= \frac{2h\mu\mu^\top}{r(z)^3}. \end{aligned} \tag{B.18}$$

For any direction $\xi \in \mathbb{R}^D$,

$$\xi^\top \nabla^2 f(z)\xi = \frac{2h}{r(z)^3}(\mu^\top \xi)^2 \ge 0.$$

Thus f is convex. Taking the maximum over the fixed feasible pairs in (7) proves that v is convex.

Together with the attained ratio representation and binary equality above, this completes the proof of Theorem 2.

C Proof of Theorem 3: global cuts and termination

We follow Algorithm 1: evaluate z^t, store its test and cut, solve the master, and then check whether another iteration is needed.

C.1 Every cut is a global lower bound

Fix iteration t and a pair (Π^t, μ^t) satisfying (6). For any $z \in [0,1]^D$, put

$$h_t = H(\Pi^t) \ge 0, \qquad r = 1 + (\mu^t)^\top z > 0, \qquad r_t = 1 + (\mu^t)^\top z^t > 0.$$

Theorem 2 gives $f_t(z) = h_t/r \le v(z)$. Since $(\mu^t)^\top(z - z^t) = r - r_t$, the cut in (8) is $L_t(z) = h_t/r_t - h_t(r - r_t)/r_t^2$. Subtracting it from the lower function gives

$$\begin{aligned} f_t(z) - L_t(z) &= h_t\left[\frac{1}{r} - \frac{1}{r_t} + \frac{r - r_t}{r_t^2}\right] \\ &= \frac{h_t[r_t^2 - rr_t + r(r - r_t)]}{rr_t^2} = \frac{h_t(r - r_t)^2}{rr_t^2} \ge 0. \end{aligned} \tag{C.1}$$

Consequently,

$$L_t(z) \le f_t(z) \le v(z) \quad (z \in [0,1]^D), \qquad L_t(z) \le V(z) \quad (z \in \mathcal{Z}_d). \tag{C.2}$$

The second inequality uses binary equality in Theorem 2. The cut is therefore valid for every allowed subset, not only for the subset where it was generated. If the pair is optimal at z^t, then

$$L_t(z^t) = f_t(z^t) = v(z^t) = V(z^t). \tag{C.3}$$

Feasibility alone proves validity; optimality is what proves touching. When $h_t = 0$, the remainder and the cut are both zero. Replacing h_t by a verified value in $[0, H(\Pi^t)]$ also preserves validity.

C.2 Bounds after every iteration

After evaluating $z^0, \dots, z^t$, let b_s be the risk upper bound of the retained original feasible test at z^s. Define

$$\begin{aligned} m_t(z) &= \max\{0, L_0(z), \dots, L_t(z)\}, \\ \mathrm{LB}_t &= \min_{z \in \mathcal{Z}_d} m_t(z), \qquad \mathrm{UB}_t = \min\{B, b_0, \dots, b_t\}. \end{aligned} \tag{C.4}$$

For a fixed z, the smallest feasible η in (9) is exactly $m_t(z)$, so LB_t is the master optimum. Equation (C.2) and $V \geq 0$ imply $m_t(z) \leq V(z)$. Every feasible test satisfies $\mathrm{OPT} \leq V(z^s) \leq b_s$; the constant test also satisfies $\mathrm{OPT} \leq B$. Thus

$$\mathrm{LB}_t = \min_{z \in \mathcal{Z}_d} m_t(z) \leq \min_{z \in \mathcal{Z}_d} V(z) = \mathrm{OPT} \leq \mathrm{UB}_t. \tag{C.5}$$

Keeping all cuts and risk bounds gives

$$m_{t+1}(z) \geq m_t(z), \qquad \mathrm{LB}_{t+1} \geq \mathrm{LB}_t, \qquad \mathrm{UB}_{t+1} \leq \mathrm{UB}_t. \tag{C.6}$$

The gap is therefore nonincreasing. Whenever it is at most ε,

$$0 \leq \mathrm{UB}_t - \mathrm{OPT} \leq \mathrm{UB}_t - \mathrm{LB}_t \leq \varepsilon. \tag{C.7}$$

The returned test has risk at most UB_t, so it has risk at most $\mathrm{OPT} + \varepsilon$.

C.3 Why exact solves terminate in finitely many steps

Assume exact continuous solutions, optimal dual pairs, globally solved masters, and no premature budget stop. At the optimal primal solution, binary equivalence gives

$$V(z^s) \leq J(a^s, \lambda^s) \leq \eta^s = v(z^s) = V(z^s).$$

Hence $b_s = J(a^s, \lambda^s) = V(z^s)$ is the bound of the original feasible test constructed by Theorem 1. Equation (C.3) gives $L_s(z^s) = V(z^s)$.

Let z^{t+1} be a global master minimizer after iteration t. If it equals an evaluated subset z^s, $s \leq t$, then

$$\begin{aligned} \mathrm{LB}_t &= m_t(z^{t+1}) = m_t(z^s) \\ &\geq L_s(z^s) = V(z^s) = b_s \geq \mathrm{UB}_t. \end{aligned} \tag{C.8}$$

Combining this with (C.5) forces $\mathrm{LB}_t = \mathrm{OPT} = \mathrm{UB}_t$. The next while-condition is false for every $\varepsilon \geq 0$, so the repeated candidate is not evaluated again. Hence

$$\mathrm{UB}_t - \mathrm{LB}_t > \varepsilon \quad \Longrightarrow \quad z^{t+1} \notin \{z^0, \dots, z^t\}.$$

Every further evaluation uses a new subset. There are only $|\mathcal{Z}_d|$ subsets, so at most $|\mathcal{Z}_d|$ continuous evaluations are possible. For $\varepsilon = 0$, the final bounds equal OPT. This proves all parts of Theorem 3.

C.4 A sufficient accuracy condition for inexact solves

Validity of the interval does not require optimal continuous solutions. If an interrupted master supplies a verified global lower bound $\ell_t \leq \min_z m_t(z)$, use

$$\mathrm{LB}_t = \max\{0, \ell_0, \dots, \ell_t\}$$

instead of the exact minimum in (C.4). Accumulated models increase, so every earlier ℓ_s remains a valid lower bound; the interval and stopping guarantee still hold. A master candidate's objective is an upper bound on its own master optimum and cannot replace ℓ_t.

For a finite-termination guarantee with inexact solves, take $\varepsilon > 0$ and nonnegative accuracy levels δ, ρ with $\delta + \rho \le \varepsilon$. Require, at every evaluation and master solve,

$$0 \le b_s - L_s(z^s) \le \delta, \qquad 0 \le m_t(z^{t+1}) - \ell_t \le \rho. \tag{C.9}$$

The first difference is the verified primal-to-cut gap at the evaluated subset; the second is the master candidate-to-bound gap. If the next candidate repeats z^s, then $m_t(z^s) \ge L_s(z^s)$, $\mathrm{UB}_t \le b_s$ and $\mathrm{LB}_t \ge \ell_t$ give

$$\begin{aligned} \mathrm{UB}_t - \mathrm{LB}_t &\le b_s - \ell_t \\ &= [b_s - L_s(z^s)] + [L_s(z^s) - m_t(z^s)] + [m_t(z^s) - \ell_t] \\ &\le \delta + 0 + \rho \le \varepsilon. \end{aligned} \tag{C.10}$$

Thus the same distinct-subset argument applies when (C.9) holds and the run is not stopped early. Merely feasible cuts, without this accuracy condition, retain valid bounds but need not make the gap converge to zero.

D Loss formulas for Section 6

Write $F(s) = p_0\Phi(-s) + p_1\Phi(s)$, so $\Psi(p_0, p_1) = \inf_s F(s)$. First take $p_0, p_1 > 0$; the axes are treated below. When the minimum is attained, s^* denotes a minimizing scalar score.

For half hinge, $\Phi(s) = (1+s)_+/2$. If $s > 1$, replacing it by 1 keeps $\Phi(-s) = 0$ and reduces $\Phi(s)$. If $s < -1$, replacing it by -1 keeps $\Phi(s) = 0$ and reduces $\Phi(-s)$. It is therefore sufficient to minimize on $[-1, 1]$:

$$\begin{aligned} F(s) &= \frac{p_0(1-s) + p_1(1+s)}{2} = \frac{p_0 + p_1}{2} + \frac{p_1 - p_0}{2}s, \\ \Psi(p_0, p_1) &= \begin{cases} F(-1) = p_0, & p_1 \ge p_0, \\ F(1) = p_1, & p_1 < p_0, \end{cases} = \min(p_0, p_1). \end{aligned} \tag{D.1}$$

The choice of endpoint follows from the sign of the linear coefficient; when $p_0 = p_1$, every score in the interval is optimal.

For logistic, $\Phi(s) = \log(1 + e^s)$. Differentiate the two terms separately:

$$\begin{aligned} \frac{d}{ds}\left[p_0 \log(1 + e^{-s})\right] &= -\frac{p_0 e^{-s}}{1 + e^{-s}} = -\frac{p_0}{1 + e^s}, \\ \frac{d}{ds}\left[p_1 \log(1 + e^{s})\right] &= \frac{p_1 e^{s}}{1 + e^{s}}, \\ F'(s) = \frac{-p_0 + p_1 e^s}{1 + e^s} = 0 \quad &\Leftrightarrow \quad -p_0 + p_1 e^{s^*} = 0 \\ &\Leftrightarrow \quad e^{s^*} = p_0/p_1, \quad s^* = \log(p_0/p_1). \end{aligned}$$

The derivative changes from negative to positive at s^*, so this is the minimum. Substituting both logarithms gives

$$\begin{aligned} 1 + e^{-s^*} &= (p_0 + p_1)/p_0, \qquad & 1 + e^{s^*} &= (p_0 + p_1)/p_1, \\ \Psi(p_0, p_1) &= p_0 \log \frac{p_0 + p_1}{p_0} + p_1 \log \frac{p_0 + p_1}{p_1} \\ &= p_0[\log(p_0 + p_1) - \log p_0] + p_1[\log(p_0 + p_1) - \log p_1] \\ &= (p_0 + p_1)\log(p_0 + p_1) - p_0 \log p_0 - p_1 \log p_1. \end{aligned} \tag{D.2}$$

For exponential, $\Phi(s) = e^s$. A square proves the lower bound and identifies equality:

$$\begin{aligned} F(s) - 2\sqrt{p_0 p_1} &= \left(\sqrt{p_0}e^{-s/2} - \sqrt{p_1}e^{s/2}\right)^2 \geq 0, \\ \sqrt{p_0}e^{-s^*/2} = \sqrt{p_1}e^{s^*/2} \quad &\Leftrightarrow \quad e^{s^*} = \sqrt{p_0/p_1}, \quad s^* = \tfrac{1}{2}\log(p_0/p_1), \\ \Psi(p_0, p_1) &= F(s^*) = 2\sqrt{p_0 p_1}. \end{aligned} \tag{D.3}$$

For squared hinge, $\Phi(s) = (1+s)_+^2$. The same clipping argument restricts the score to $[-1, 1]$. Expanding and completing the square gives

$$\begin{aligned} F(s) &= p_0(1-s)^2 + p_1(1+s)^2 \\ &= (p_0 + p_1)s^2 + 2(p_1 - p_0)s + (p_0 + p_1) \\ &= (p_0 + p_1)\left(s - \frac{p_0 - p_1}{p_0 + p_1}\right)^2 + (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1}, \\ (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1} &= \frac{(p_0 + p_1)^2 - (p_0 - p_1)^2}{p_0 + p_1} = \frac{4p_0 p_1}{p_0 + p_1}, \\ s^* = \frac{p_0 - p_1}{p_0 + p_1} \in [-1, 1], \qquad &\Psi(p_0, p_1) = F(s^*) = \frac{4p_0 p_1}{p_0 + p_1}. \end{aligned} \tag{D.4}$$

The nonnegative square vanishes at this feasible score, which proves the minimum.

On the axes, nonnegativity and the indicated score limits give

$$\begin{aligned} 0 \leq \Psi(p_0, 0) \leq p_0\Phi(-s) \longrightarrow 0 \quad (s \to +\infty), \\ 0 \leq \Psi(0, p_1) \leq p_1\Phi(s) \longrightarrow 0 \quad (s \to -\infty). \end{aligned}$$

Consequently,

$$\Psi(p_0, 0) = \lim_{s \to +\infty} p_0\Phi(-s) = 0, \qquad \Psi(0, p_1) = \lim_{s \to -\infty} p_1\Phi(s) = 0, \qquad \Psi(0, 0) = 0.$$

For half hinge and squared hinge, the first zero is already attained at $s \geq 1$, and the second at $s \leq -1$. For logistic and exponential with a positive remaining mass, it is only an infimum as the score tends in the stated direction. The closed formulas use $0 \log 0 = 0$ and assign the squared-hinge value zero when both masses are zero.